INFOSYS SE PSEUDOCODE VAULT • ARRAYS, STRINGS & DATA STRUCTURES

1D/2D Arrays, Matrices, Element Swapping, String Indexing • Total Solved Questions: 41

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Index Calculation

Q1. What will be the output of the following pseudocode?

```
Set Integer a = [4, 7, 2, 9, 5]
display a[a[2]] + a[1]
```

(A) 7
(B) 9
(C) 11
(D) 13

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** a[2]=2, so a[a[2]]=a[2]=2; 2+7=9.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Transformation

Q2. What will be the output of the following pseudocode?

```
Set Integer a = [3, 5, 7, 9]
for i = 0 to 3 step 1 do
a[i] = a[i] + i
end-for
display a[3] - a[1]
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (D) : 6

■ **Step-by-Step Logic:** After updates, a=[3,6,9,12]; 12−6=6.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Maximum

Q3. What will be the output of the following pseudocode?

```
Set Integer a = [8, 3, 15, 6, 11]
Set Integer m = a[0]
for i = 1 to 4 step 1 do
if (a[i] > m)
m = a[i]
end-if
end-for
display m % 4
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (C) : 3

■ **Step-by-Step Logic:** Maximum is 15; 15%4=3.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Pair Sum

Q4. What will be the output of the following pseudocode?

```
Set Integer a = [2, 5, 8, 11]
Set Integer s = 0
for i = 0 to 2 step 1 do
s = s + a[i] + a[i+1]
end-for
display s
```

(A) 36
(B) 39
(C) 42
(D) 45

Correct Answer: Option (B) : 39

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (B) (39).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Conditional Update

Q5. What will be the output of the following pseudocode?

```
Set Integer a = [4, 7, 10, 13]
for i = 0 to 3 step 1 do
if (a[i] % 2 == 0)
a[i] = a[i] / 2
else
a[i] = a[i] + 1
end-if
end-for
display a[0] + a[1] + a[2] + a[3]
```

(A) 25
(B) 27
(C) 29
(D) 31

Correct Answer: Option (C) : 29

■ **Step-by-Step Logic:** Array becomes [2,8,5,14]; sum=29.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Reverse

Q6. What will be the output of the following pseudocode?

```
Set Integer a = [3, 6, 9, 12, 15]
display a[4] - a[0] + a[3] - a[1]
```

(A) 12
(B) 15
(C) 18
(D) 21

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** 15−3+12−6=18. Infosys Medium Pseudocode Practice • Page 39

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Count

Q7. What will be the output of the following pseudocode?

```
Set Integer a = [5, 10, 15, 20, 25]
Set Integer c = 0
for i = 0 to 4 step 1 do
if (a[i] % 10 == 5)
c = c + 1
end-if
end-for
display c
```

(A) 2
(B) 3
(C) 4
(D) 5

Correct Answer: Option (B) : 3

■ **Step-by-Step Logic:** 5,15,25 satisfy the condition: 3.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Running Sum

Q8. What will be the output of the following pseudocode?

```
Set Integer a = [2, 4, 6, 8]
Set Integer s = a[0]
for i = 1 to 3 step 1 do
a[i] = a[i] + s
s = a[i]
end-for
display a[3]
```

(A) 16
(B) 20
(C) 24
(D) 30

Correct Answer: Option (B) : 20

■ **Step-by-Step Logic:** Running values become 2,6,12,20; a[3]=20.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Nested Index

Q9. What will be the output of the following pseudocode?

```
Set Integer a = [1, 3, 0, 2]
display a[a[3]] + a[a[1]]
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (B) : 2

■ **Step-by-Step Logic:** a[3]=2, so a[2]=0; a[1]=3, so a[3]=2; total=2.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Difference

Q10. What will be the output of the following pseudocode?

```
Set Integer a = [12, 5, 9, 4]
Set Integer m = a[0]
for i = 1 to 3 step 1 do
if (a[i] < m)
m = a[i]
end-if
end-for
display a[0] - m
```

(A) 6
(B) 7
(C) 8
(D) 9

Correct Answer: Option (C) : 8

■ **Step-by-Step Logic:** Minimum is 4; 12−4=8.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Shift

Q11. What will be the output of the following pseudocode?

```
Set Integer a = [3, 5, 7, 9]
display (a[3] >> 1) + (a[0] << 1)
```

(A) 8
(B) 10
(C) 12
(D) 14

Correct Answer: Option (B) : 10

■ **Step-by-Step Logic:** 9>>1=4; 3<<1=6; total=10.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array XOR

Q12. What will be the output of the following pseudocode?

```
Set Integer a = [6, 10, 12, 3]
display (a[0] ^ a[1]) + a[3]
```

(A) 13
(B) 15
(C) 17
(D) 19

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (15).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Pair Comparison

Q13. What will be the output of the following pseudocode?

```
Set Integer a = [4, 9, 6, 11]
Set Integer c = 0
for i = 0 to 2 step 1 do
if (a[i] < a[i+1])
c = c + 1
end-if
end-for
display c
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (B) : 2

■ **Step-by-Step Logic:** The true comparisons are 4<9 and 6<11, so count=2.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Sum with Step

Q14. What will be the output of the following pseudocode?

```
Set Integer a = [2, 4, 6, 8, 10, 12]
Set Integer s = 0
for i = 0 to 5 step 2 do
s = s + a[i]
end-for
display s
```

(A) 12
(B) 16
(C) 18
(D) 20

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** Indices 0,2,4 give 2+6+10=18. Infosys Medium Pseudocode Practice • Page 40

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Rotation Logic

Q15. What will be the output of the following pseudocode?

```
Set Integer a = [10, 20, 30, 40]
Set Integer t = a[3]
for i = 3 to 1 step -1 do
a[i] = a[i-1]
end-for
a[0] = t
display a[1] + a[3]
```

(A) 30
(B) 40
(C) 50
(D) 60

Correct Answer: Option (B) : 40

■ **Step-by-Step Logic:** After right rotation: [40,10,20,30]; a[1]+a[3]=40.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Length

Q16. What will be the output of the following pseudocode?

```
Set String s = "PLACEMENT"
Set Integer n = stringLength(s)
display (n << 1) - 3
```

(A) 13
(B) 15
(C) 17
(D) 19

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** Length=9; 9<<1=18; 18−3=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Indexing

Q17. What will be the output of the following pseudocode?

```
Set String s = "INFOSYS"
display s[1] + s[5]
```

(A) NS
(B) NY
(C) FY
(D) SY

Correct Answer: Option (B) : NY

■ **Step-by-Step Logic:** s[1]=N and s[5]=Y, so the result is NY.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Condition

Q18. What will be the output of the following pseudocode?

```
Set String s = "COMPUTER"
if (stringLength(s) % 2 == 0)
display stringLength(s) / 2
else
display stringLength(s) + 2
end-if
```

(A) 4
(B) 5
(C) 6
(D) 10

Correct Answer: Option (A) : 4

■ **Step-by-Step Logic:** COMPUTER has 8 characters; 8/2=4.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String + Loop

Q19. What will be the output of the following pseudocode?

```
Set String s = "CODE"
Set Integer total = 0
for i = 0 to stringLength(s)-1 step 1 do
total = total + i
end-for
display total
```

(A) 4
(B) 5
(C) 6
(D) 7

Correct Answer: Option (C) : 6

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (C) (6).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Index + Shift

Q20. What will be the output of the following pseudocode?

```
Set String s = "TESTING"
Set Integer n = s[1] == 'E'
display stringLength(s) << n
```

(A) 7
(B) 14
(C) 21
(D) 28

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** The comparison is true, represented as 1; length 7<<1=14.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Length + Modulo

Q21. What will be the output of the following pseudocode?

```
Set String s = "ALGORITHM"
display stringLength(s) % 4
```

(A) 0
(B) 1
(C) 2
(D) 3

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** ALGORITHM has 9 characters; 9%4=1.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Reverse Access

Q22. What will be the output of the following pseudocode?

```
Set String s = "INFOSYS"
display s[stringLength(s)-1]
```

(A) I
(B) S
(C) Y
(D) Error

Correct Answer: Option (B) : S

■ **Step-by-Step Logic:** Last index is 6; s[6] = S. Infosys Medium Pseudocode Practice • Page 41

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Character Count

Q23. What will be the output of the following pseudocode?

```
Set String s = "PLACEMENT"
Set Integer c = 0
for i = 0 to stringLength(s)-1 step 1 do
if (s[i] == 'E')
c = c + 1
end-if
end-for
display c
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (B) : 2

■ **Step-by-Step Logic:** PLACEMENT contains E twice.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Length + Right Shift

Q24. What will be the output of the following pseudocode?

```
Set String s = "INTERVIEW"
display stringLength(s) >> 1
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (B) : 4

■ **Step-by-Step Logic:** INTERVIEW has 9 characters; 9>>1=4.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Conditional

Q25. What will be the output of the following pseudocode?

```
Set String s = "NQT2026"
if (stringLength(s) > 6)
display s[0]
else
display s[1]
end-if
```

(A) N
(B) Q
(C) T
(D) 2

Correct Answer: Option (A) : N

■ **Step-by-Step Logic:** Length is 7, so the first branch displays s[0]=N.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Array Formula

Q26. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b, Integer c)
return (a + b) ^ c
End function
display f(5, 3, 6)
```

(A) 4
(B) 8
(C) 12
(D) 14

Correct Answer: Option (D) : 14

■ **Step-by-Step Logic:** 5+3=8; 8 XOR 6 = 14.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Array + Function

Q27. What will be the output of the following pseudocode?

```
Integer f(Integer x)
return (x >> 1) + 3
End function
Set Integer a = [8, 12, 16]
display f(a[2])
```

(A) 9
(B) 10
(C) 11
(D) 12

Correct Answer: Option (C) : 11

■ **Step-by-Step Logic:** a[2]=16; 16>>1=8; 8+3=11.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Array + Right Shift

Q28. What will be the output of the following pseudocode?

```
Set Integer a = [18, 25, 32, 41]
display (a[3] >> 2) + (a[0] % 5)
```

(A) 11
(B) 12
(C) 13
(D) 14

Correct Answer: Option (C) : 13

■ **Step-by-Step Logic:** 41>>2=10; 18%5=3; total=13. Infosys Medium Pseudocode Practice • Page 44

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Loop + Array Update

Q29. What will be the output of the following pseudocode?

```
Set Integer a = [1, 2, 3, 4, 5]
for i = 1 to 4 step 1 do
a[i] = a[i] + a[i-1]
end-for
display a[4]
```

(A) 10
(B) 12
(C) 15
(D) 16

Correct Answer: Option (C) : 15

■ **Step-by-Step Logic:** Running cumulative values become [1,3,6,10,15].

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q30. What will be the output of the following pseudocode?

```
[Arrays]
What is printed?
Integer
A[4] = {3, 5, 7, 9}
Integer
i, p
Set
p = 1
for
i = 0 to 3 do
p
= p * A[i]
end
for
Print
p
```

(A) 24
(B) 315
(C) 945
(D) 105

Correct Answer: Option (C) : 945

■ **Step-by-Step Logic:** Product 3×5×7×9 = 945.

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q31. What will be the output of the following pseudocode?

```
1.     Which of the following is a valid declaration of a
char?
```

(A) char ch = '\utea';
(B) char ca = 'tea';
(C) char cr = \u0223;
(D) char cc = '\itea';

Correct Answer: Option (A) : char ch = '\utea';

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (char ch = '\utea';).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q32. 1. What will be the output of this program fragment?

```
int i = 263;
      putchar(i);
```

(A) prints 263
(B) prints the ASCII equivalent of 263
(C) rings the bell
(D) prints garbage

Correct Answer: Option (C) : rings the bell

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (C) (rings the bell).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q33. What will be the output of the following pseudocode?

```
1.     Let A be a square matrix of size nxn. Consider the
following program and state the output.
C=100
for i= 1 to n do
for j = 1 to n do
{
Temp = A[i][j] + C
A[i][j] = A [j][i]
A[j][i]= Temp – C
}for i = 1 to n do
for j = 1 to n do

Output (A[i][j]);
```

(A) a) The matrix A itself
(B) a) Transpose of matrix A
(C) a) Adding 100 to the upper diagonal elements and subtracting 100 from diagonal elements of A
(D) a) None of the above

Correct Answer: Option (A) : a) The matrix A itself

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (A) (a) The matrix A itself).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q34. What will be the output of the following pseudocode?

```
1. Integer a = 5, b = 9 2. a = a ^ b 3. b = a ^ b 4. a = a
^ b 5. if (a == 9 && b == 5) 6. Print "Swapped" 7. else
8. Print "Not swapped"
```

(A) Line 2
(B) Line 3
(C) Line 4
(D) No error

Correct Answer: Option (D) : No error

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (No error).

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q35. What will be the output of the following pseudocode?

```
What is the final content of arr? (indexing starts at 0;
updates are in place)
arr = [4, 1, 3, 2]
for i = 0 to 3
        arr[i] = arr[i] + arr[arr[i] % 4]
end for
```

(A) [8, 2, 5, 7]
(B) [8, 2, 5, 6]
(C) [8, 5, 5, 7]
(D) [5, 2, 5, 7]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 2]

Topic: Infosys SE College Mock Test

Q36. What will be the output of the following pseudocode?

```
What is the final content of arr? (0-indexed; each
statement uses the array's state at that point)
arr = [1, 2, 3, 4, 5]
temp = arr[0]
arr[0] = arr[4]
arr[4] = arr[2]
arr[2] = arr[1]
arr[1] = temp
```

(A) [5, 1, 2, 4, 3]
(B) [5, 1, 3, 4, 2]
(C) [5, 2, 1, 4, 3]
(D) [5, 1, 2, 4, 2]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 3]

Topic: Infosys SE College Mock Test

Q37. What will be the output of the following pseudocode?

```
What is the final content of arr? (0-indexed; each
statement uses the array's state at that point)
```

(A) [60, 40, 30, 20, 50, 10]
(B) [60, 20, 30, 40, 50, 10]
(C) [10, 40, 30, 20, 50, 60]
(D) [60, 40, 50, 20, 30, 10]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 4]

Topic: Infosys SE College Mock Test

Q38. How many swaps are performed in total?

```
arr = [5, 2, 8, 1, 9, 3]
swaps = 0
for i = 0 to 4
        for j = 0 to 4 - i
                if (arr[j] > arr[j+1])
                        temp = arr[j]
                        arr[j] = arr[j+1]
                        arr[j+1] = temp
                        swaps = swaps + 1
                end if
        end for
end for
print(swaps)
```

(A) 7
(B) 8
(C) 6
(D) 9

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 4]

Topic: *Infosys SE College Mock Test*

Q39. What will be the output of the following pseudocode?

```
This algorithm moves all non-zero elements to the front,
preserving order, and fills the rest with zeroes.
What is the final arr?
arr = [3, 0, 4, 0, 5, 0, 2]
pos = 0
for i = 0 to 6
        if (arr[i] != 0)
                arr[pos] = arr[i]
                pos = pos + 1
        end if
end for
while (pos <= 6)
        arr[pos] = 0
        pos = pos + 1
end while
print(arr)
```

(A) [3, 4, 5, 2, 0, 0, 0]
(B) [3, 0, 4, 0, 5, 0, 2]
(C) [3, 4, 5, 2, 0, 0, 2]
(D) [0, 0, 0, 3, 4, 5, 2]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 5]

Topic: *Infosys SE College Mock Test*

Q40. This is a binary search for target = 50 in a sorted array. How many comparisons are made before

```
the loop ends (the target is not present)?
arr = [2, 5, 8, 12, 16, 23, 38, 45, 56, 72, 91]
low = 0; high = 10; target = 50; comparisons = 0
while (low <= high)
        mid = (low + high) / 2
        comparisons = comparisons + 1
        if (arr[mid] == target)
                break
        else if (arr[mid] < target)
                low = mid + 1
        else
                high = mid - 1
        end if
end while
print(comparisons)
```

(A) 4
(B) 3
(C) 5
(D) 6

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 5]

Topic: *Infosys SE College Mock Test*

Q41. What will be the output of the following pseudocode?

```
This is Kadane's algorithm for maximum subarray sum. What
is printed?
arr = [-2, 3, -1, 5, -3, 4, -2, 6]
maxSoFar = arr[0]; maxEndingHere = arr[0]
for i = 1 to 7
        maxEndingHere = max(maxEndingHere + arr[i],
arr[i])
        maxSoFar = max(maxSoFar, maxEndingHere)
end for
print(maxSoFar)
```

(A) 12
(B) 10
(C) 9
(D) 15

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .